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REVIEW

Rapid in-field soil analysis of plant-available nutrients and pH for precision agriculture—a review

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Abstract

There are currently many in-field methods for estimating soil properties (e.g., pH, texture, total C, total N) available in precision agriculture, but each have their own level of suitability and only a few can be used for direct determination of plant-available nutrients. As promising approaches for reliable in-field use, this review provides an overview of electromagnetic, conductivity-based, and electrochemical techniques for estimating plant-available soil nutrients and pH. Soil spectroscopy, conductivity, and ion-specific electrodes have received the most attention in proximal soil sensing as basic tools for precision agriculture during the last two decades. Spectral soil sensors provide indication of plant-available nutrients and pH, and electrochemical sensors provide highly accurate nitrate and pH measurements. This is currently the best way to accurately measure plant-available phosphorus and potassium, followed by spectral analysis. For economic and practicability reasons, the combination of multi-sensor in-field methods and soil data fusion has proven highly successful for assessing the status of plant-available nutrients in soil for precision agriculture. Simultaneous operation of sensors can cause problems for example because of mutual influences of different signals (electrical or mechanical). Data management systems provide relatively fast availability of information for evaluation of soil properties and their distribution in the field. For rapid and broad adoption of in-field soil analyses in farming practice, in addition to accuracy of fertilizer recommendations, certification as an official soil analysis method is indispensable. This would strongly increase acceptance of this innovative technology by farmers.

Keywords Proximal soil sensing · Real-time · Sensing technologies · Sensor fusion · Precision farming

Introduction

Knowledge of soil properties such as soil texture, organic matter, pH, and the current status of plant-available nutrients is required for precise fertilizer recommendations for arable soils (Jordan-Meille et al., 2012). This knowledge is acquired through soil

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analysis, which is commonly done in three steps: soil sampling in the field followed by transport to, and analysis in the laboratory.

However, conventional soil sampling and laboratory analysis cannot provide rapid information because transport and sampling logistics are time consuming. Hence, the results may not be available at the right time for fertilization, which can limit proper fertilizer management (Hinck et al., 2018). Furthermore, laboratory analyses require chemicals that can harm the environment (Guerrero et al., 2021). The number of soil samples per field or area unit is usually smaller than would be required to consider field heterogeneity (Domsch & Schirrmann, 2009). Intensive research is therefore being carried out on systems for rapid in-field analysis of plant-available soil nutrients, numerous other soil characteristics and their respective spatial distribution.

Most approaches for rapid in-field soil analysis are based on technologies like near-infrared (NIR) and mid-infrared (MIR) spectroscopy (Masrie et al., 2018; Ng et al., 2022), capillary electrophoresis (Smolka et al., 2017), satellite hyper-spectral remote sensing (Mazur et al., 2022), laser-induced breakdown spectroscopy (LIBS) (Erler et al., 2020), or.

X-ray fluorescence spectroscopy (XRF) (Tavares, 2021), but not all of them are able to detect plant-available nutrient fractions. Moreover, whilst proof of principle has already been successfully provided for several of these technical approaches (Metzger et al., 2024; Piccini et al., 2024), their successful application under practical agricultural conditions in the field has yet to be demonstrated.

In the context of measuring soil parameters in the field, remote and proximal soil sensing approaches are distinguished (e.g. Piccini et al., 2024). In the modern satellite era, it is possible to acquire soil information from remote sensing data (e.g., from the Sentinel-1 or Sentinel-2 satellites of the European Space Agency). Information on important parameters such as soil texture and humus content can be derived from satellite data with a high degree of reliability (Mazur et al., 2022; Toscano et al., 2019; Yuzugullu et al., 2024). On the other hand, proximal soil sensing refers to field-based techniques that can measure soil properties from 2 m or less above the soil surface (Adamchuk et al., 2010, 2018; Behera et al., 2022; Lobsey et al., 2010). Furthermore, the combination of various proximal soil sensing technologies enables the compilation of more comprehensive soil information and helps to optimize the characterization of the soil (Archbold et al., 2023; Hinck et al., 2023; Magno et al., 2024; Molin & Tavares, 2019; Tabatabai et al., 2019; Viscarra Rossel et al., 2006; Xu et al., 2019). Furthermore, current introduction of machine learning approaches could complement to proximal soil sensing and lead to powerful soil analysis, including soil nutrients, among other soil properties (e.g. Islam et al., 2023; Lincy et al., 2024; Ma et al., 2023).

In the last three decades, many reviews about the sensing of soil properties for precision agriculture have been published (e.g. Adamchuk et al., 2004, 2018, 2021; Hummel et al., 1996; Nawar et al., 2017; Shibusawa, 2006; Sudduth et al., 1997; Viscarra Rossel et al., 2011). To our knowledge, there are only a limited number of published studies that focus on rapid in-field measurements focusing on plant-available nutrients and pH in arable soils that can be used directly for fertilizer recommendations. As rapid in-field methods offer great potentials, a review article on these methods is worthwhile.

This review is about in-field techniques for measuring plant-available soil nutrients and pH determined with proximal soil sensors that are currently discussed for precision agriculture. Their importance is highlighted in case studies, describing the strengths and weaknesses of the various approaches. As there is such a wealth of methods based on a wide range of analytical principles, the coverage may not be complete in all cases.

Following this introduction, this article is organized as follows: Sect. 2 summarizes information regarding principles for rapid in-field measurement of plant-available soil nutrients and pH like measurement types (2.1), sensor measurement techniques (2.2) and practical tools for in-field soil analysis (2.3), case studies for in-field analyses (2.4), and combinations of sensors (2.5). Section 3 addresses aspects of quality management including a few statistical terms for better understanding, discussion of the accuracy of sensor measurements, details of calibration and validation to ensure the reliability of the measurements. In Sect. 4 concluding remarks are presented.

Principles for in-field measurement of plant-available soil nutrients and pH

Recently, the focus of research in assessing soil properties in the field has moved to proximal sensors. The majority of these sensors operate with electromagnetic radiation (so-called spectral sensors), but electrochemical or geoelectrical resistivity sensors are also used. According to Viscarra Rossel et al. (2011), proximal soil sensing can be categorized according to several aspects: (1) invasive or non-invasive measurement, (2) active or passive energy source, (3) static or mobile mode of operation, and (4) direct or indirect derivation of soil properties. According to Archbold et al. (2023) and Guerrero et al. (2021), electrochemical and spectral sensors are favorable for measuring plant-available soil nutrients. Overall, however, it should be kept in mind that all kinds of sensors require proper calibration and validation against standard laboratory methods to ensure accurate and reliable results. An overview of some existing rapid in-field methods for soil nutrient analysis is provided in Table 1. Measurement methods for determining plant-available nutrients and pH along with their accuracies are summarized in Table 2.

Measurement types

When sensor measurements of soil samples are done in the field, regardless of whether they are based on invasive or non-invasive methods, they are considered in-situ measurements because the analysis is performed directly in the field, i.e., without any need for sample transport for laboratory analysis (Archbold et al., 2023; Kuang et al., 2012). In principle, there are two possible ways to measure soil nutrients in the field. The first relies on invasive sensors that are inserted into the soil to collect data on certain soil characteristics. These sensors are typically designed as long probes, tubes, or rods (e.g. Chrysalabs, 2023; Stenon, 2021), and include both in-situ sensors that take measurements directly in the soil, and ex-situ ones that measure excavated, disturbed soil samples (e.g. Metzger et al., 2024). The second way involves non-invasive technologies that operate without contact between the soil and sensor. The most common measurements carried out with invasive soil sensors address soil moisture, temperature, and salinity (Dwevedi et al., 2017). Some more advanced sensors (e.g. NIR, ISEs, ISFET) are also able to measure plant-available nutrients and typically include one or more sensors mounted on the probe, having wired or wireless communication capabilities to transmit the collected data (Chrysalabs, 2023; Decker et al., 2019; Gerken, 2023; Hinck et al., 2023; Sethuramasaamyraja et al., 2008; Stenon, 2021). One limitation of these invasive sensors is that they can often only record very localized data, as they only take measurements at a single, narrowly defined point in the soil. For larger heterogeneous fields, repeated measurements at different in-field locations may be required to detect the field-specific variability

Table 1 Overview of rapid in-field soil nutrient analysis tools considering key specifications

Soil analysis tool [a]Calibration required	Operating method and measurement type Static (in-/ex-situ) / machine-based / handheld	Multi Sensor Platform available	Sampling planning and geolocation with GNSS	Preparation of the sample before measuring required	pH and plant-available nutrients measured and their accuracy (see footnote)				Operation principle	Measurement methods (sensor technique specifications); Inference: direct (D)/ indirect (I)	Data Management	Automatic nutrient / fertilizer recommendation	Reference(s) and links to selected images/providers
					pH	N	P	K					
[a]FieldLab[2,3] soil2data	Machine-based, invasive (in-situ)	x	x	x	A	A	B	A	Multisensory: electrochemical	ISFET; (D)	x cloud based	x	Riedel et al. (2022); Najdenko et al. (2023) 2018-MCG-Concept_and_first_results_of_a_field-robot-based_on-the-go_assessment_of_soil_nutrients_with_ion-sensitive_field_effect_transistors.pdf (hs-osnabrueck.de)
ChrysaLabs[1]	Handheld probe, invasive (in-/ex-situ)	x	x	–	n.a	n.a	n.a	n.a	AI based analysis: spectral, gamma-ray, electrical capacitance technology	Vis–NIR + Capacitance Technology + ATR; (D/I)	x cloud based	x	https://www.chrysalabs.com/
[a]Farmlab (Stenon) [1]	Handheld probe, invasive (in-situ)	x	x	–	C	C-E	E	D	AI based analysis: spectral, geoelectrical	Spectral library Vis–NIR, EC;(D/I)	x cloud based	x	Stenon (2021); Olfs et al. (2023); Vikuk et al. (2024) https://stenon.io/en/

Table 1 (continued)

Soil analysis tool [a]Calibration required	Operating method and measurement type Static (in-/ex-situ) / machine-based / handheld	Multi Sensor Platform available	Sampling planning and geolocation with GNSS	Preparation of the sample before measuring required	pH and plant-available nutrients measured and their accuracy (see footnote)				Operation principle	Measurement methods (sensor technique specifications); Inference: direct (D)/ indirect (I)	Data Management	Automatic nutrient / fertilizer recommendation	Reference(s) and links to selected images/providers
					pH	N	P	K					
[a]AgroCares (SoilCares) [(1)]	Handheld probe, non-invasive, (in-situ)	–	–	–	x	n.a	n.a	n.a	Spectral	NIR (1400 – 2500 nm); (I)	x cloud based	SoilCare Adviser	https://agrocares.com/scanners/
[a]Veris MSP[(1)]	Machine-based, invasive (in-situ)	x	x	–	A	B	n.a	B	Multisensory: electrochemical, geoelectrical	ISEs (antimony, PVC), EC; (D/I)	x	–	Sethuramasamyraja et al. (2008); Schirrmann et al. (2011); http://www.veristech.com/the-sensors
[a]Veris MSP[(1)]	Machine-based, invasive	x	x	–	B	n.a	n.a	n.a	Multisensory: electrochemical, geoelectrical,	ISEs (glass), EC; (D/I)	x	–	Ruehlmann et al., (2021, 2023)
[a]Veris[(1)]	Machine-based, invasive	x	x	–	A	n.a	n.a	n.a	Electrochemical	ISEs (antimony); (D)	agriPORT	x	Decker et al. (2019); https://www.agricon.de/
[a]Veris[(1)]		x	–	–	B	n.a	n.a	n.a	Geoelectrical	EC	–	–	Borchert et al. (2010)

Table 1 (continued)

Soil analysis tool [a]Calibration required	Operating method and measurement type Static (in-/ex-situ) / machine-based / handheld	Multi Sensor Platform available	Sampling planning and geolocation with GNSS	Preparation of the sample before measuring required	pH and plant-available nutrients measured and their accuracy (see footnote)				Operation principle	Measurement methods (sensor technique specifications); Inference: direct (D)/ indirect (I)	Data Management	Automatic nutrient / fertilizer recommendation	Reference(s) and links to selected images/providers
					pH	N	P	K					
[a]Veris[(2)]	Machine-based, invasive (in-situ)	x	x	x	A	B	n.a	C	Multisensory: electrochemical, geoelectrical	ISEs (glass, PVC), EC; (D/I)			Adamchuk et al. (2005)
[a]Veris MSP3™[(2)]	Machine-based, invasive / non-invasive	x	x	–	A	n.a	n.a	n.a	Multisensory: spectral, electrochemical, geoelectrical	NIR (660–940 nm), ISEs, EC; (D/I)	x	–	Kweon et al., (2012, 2013) Veris MSP3 Soil Scanner—Vantage Agrometius (vantage-agrometius.nl)
[(2,3)]stationary in-situ sensor system	Static, invasive (in-situ)	–	–	x	n.a	A	B	B	Electrochemical	ISEs (PVC); (D)	–	–	Kim et al. (2013)
[a]Veris MSP[(1)]	Machine-based, invasive (in-situ)	x	x	–	A	B	n.a	B	Multisensory: electrochemical, geoelectrical	ISEs (antimony, PVC), EC; (D/I)	x	–	Sethuramasamyraja et al. (2008); Schirrmann et al. (2011); http://www.veristech.com/the-sensors

Table 1 (continued)

Soil analysis tool [a]Calibration required	Operating method and measurement type Static (in-/ex-situ) / machine-based / handheld	Multi Sensor Platform available	Sampling planning and geolocation with GNSS	Preparation of the sample before measuring required	pH and plant-available nutrients measured and their accuracy (see footnote)				Operation principle	Measurement methods (sensor technique specifications); Inference: direct (D)/ indirect (I)	Data Management	Automatic nutrient / fertilizer recommendation	Reference(s) and links to selected images/providers
					pH	N	P	K					
[a]Real-time soil analysis system [2,3]	Invasive (in-situ)	x	–	x	n.a	A	n.a	n.a	Multisensory: electrochemical	ISFET/ FIA; (D)	–	–	Birrel an Hummel (2001)
SNMS	Machine-based/ static, handheld	x	x	x	n.a	A	n.a	n.a	Electrochemical	ISEs (PVC); (D)	x	–	Sibley et al. (2010) (3) (PDF) Field-scale validation of an automated soil nitrate extraction and measurement system (researchgate.net)
X-ray tube[2]	In-situ, non-invasive	–	–	x	D	n.a	A	D	Multisensory: spectral. Spectral library	Vis–NIR, XRF, LIBS; (I)	x	n.a	Tavares et al., (2020, 2021)
Easy-to-use versatile nutrients analysis tools[2]	In-situ, non-invasive	–	–	x	n.a	n.a	n.a	A	Spectral	XRF; (I)	–	–	Tavares et al. (2023) 1-s2.0-S0016706123003786-gr1_lrg.jpg (1711×885) (els-cdn.com)

Table 1 (continued)

Soil analysis tool [a]Calibration required	Operating method and measurement type Static (in-/ex-situ) / machine-based / handheld	Multi Sensor Platform available	Sampling planning and geolocation with GNSS	Preparation of the sample before measuring required	pH and plant-available nutrients measured and their accuracy (see footnote)				Operation principle	Measurement methods (sensor technique specifications); Inference: direct (D)/ indirect (I)	Data Management	Automatic nutrient / fertilizer recommendation	Reference(s) and links to selected images/providers
					pH	N	P	K					
[a]I4S RapidMapper[2,3]	Machine-based, non-invasive	x	x	–	A	n.a	B (Vis–NIR, MIR)	n.a	Multisensory: spectral, electrochemical, geoelectrical. Spectral library	Vis–NIR (500–2400 nm), NIR (860–2550 nm), MIR (2500–16667 nm), XRF, LIBS, ISEs, EC; (D/I)	DSS	In plan, but exist liming and irrigation recommendations	Pätzold et al. (2020); Tavakoli et al. (2022) Sensor-Plattform zum ortsspezifischen Bodenmanagement | Bioökonomie.de (biooekonomie.de) I4S: Results (atb-potsdam.de)
"Soilmonitor"[2]	Static, invasive	–	–	x	n.a	n.a	n.a	n.a	Spectral, electrochemical, microfluidic sensor system	optoelectronic detection chip with organic light-emitting diodes; (D)	fusion with remote sensing (satellite or drohnen)	planned	Gerken, 2023 SOILMONITOR (zevs-kiel.de) 173-Soilmonitor (uni-kiel.de)
NutriTest sensor[1]	Handheld	–	–	x	n.a	In accurate manner			AI based analysis: spectral (colorimetric sensing)	LED, LDR; (D)	x	–	Lincy et al. (2024)

Table 1 (continued)

Soil analysis tool [a]Calibration required	Operating method and measurement type Static (in-/ex-situ) / machine-based / handheld	Multi Sensor Platform available	Sampling planning and geolocation with GNSS	Preparation of the sample before measuring required	pH and plant-available nutrients measured and their accuracy (see footnote)				Operation principle	Measurement methods (sensor technique specifications); Inference: direct (D)/ indirect (I)	Data Management	Automatic nutrient / fertilizer recommendation	Reference(s) and links to selected images/providers
					pH	N	P	K					
[a]NeoSpectra (NEO) MEMS based spectrometer[(1)]	Handheld, non-invasive, (in-situ)	–	–	x	D	n.a	A	C	Spectral	FT-NIR (1350–2500 nm); (I)	–	–	Metzger et el. (2024)
[a]PRS research-grade Spectral Evolution spectrophotometer[(2)]	Handheld, non-invasive, (in-situ)	–	–	x	B	n.a	A	C	Spectral	Vis–NIR (350-2100 nm); (I)	–	–	Metzger et el. (2024)

Categorization of the technologies in terms of availability for in-field application: commercially available[(1)], under development[(2)] or prototypes available[(3)]

Information about the need for calibration is given in column 1

The scale of accuracy (Chang et al., 2001; Kuang et al., 2012; Lorenz et al., 2023) is indicated by the letter A–E, as follows: A, excellent ($R^2 \geq 0.8$)—measurable with very high accuracy; B, good ($0.6 \geq R^2 < 0.8$)—measurable with high accuracy; C, approximate quantitative prediction ($0.4 \geq R^2 < 0.6$)—measurable with medium accuracy; D, distinguish between high and low ($0.2 \geq R^2 < 0.4$)—measurable with low accuracy; E, not usable ($R^2 < 0.02$)—measurable with very low accuracy; and n.a. indicates no data, not measurable, or not mentioned in the literature.

Other abbreviations are as follows: *N* nitrate, *P* phosphate, *K* potassium, *GNSS* Global Navigation Satellite System, *SNMS* Soil Nitrate Mapping System, *AI* machine-learning technologies for artificial intelligence analysis, *ATR* attenuated total reflection (a measurement technique used in conjunction with infrared spectroscopy that enables samples to be examined directly in the solid or liquid state without further preparation), *FIA* flow injection analysis, *Vis–NIR* visible near-infrared, *XRF* X-ray fluorescence, *ISEs* ion-selective electrodes, *PVC* polyvinyl chloride membrane for ISEs for measuring macronutrients, *ISFET* ion-selective field effect transistors, *LIBS* laser-induced breakdown spectroscopy, *EC* geoelectrical conductivity, *I4S* Intelligence for Soil, *DSS* decision support system, *LED* light-emitting diode used as a light source, *LDR* Light Dependent Resistor used as light sensors, *MEMS* micro-electro-mechanical systems, *FT-NIR* handheld Fourier-transform near-infrared. Explanations of other terms: Invasive in-situ measurement—takes measurements in the soil or soil extract; invasive ex-situ measurement—takes measurements in excavated soil; static operation—requires stationary measurements; handheld operation—easily transportable device or tool that can be carry by hand; machine-based operation— self-propelled or non-self-propelled tool that can moved independently or be attached to a vehicle; direct inference—measures a physical process or property; indirect inference—measures a proxy process or property

[a]Requires calibration on laboratory samples at the site-specific, farm-specific, or regional level; "x" = available; "- " = no information found

Table 2 Overview of plant-available soil nutrients and pH measured with some proximal soil sensors along with their measurement accuracy

Sensor technology		R^2	Accuracy[(a)]	Reference(s)
pH				
Spectral	MIR	0.67	B	Janik et al. (1998);
		0.72	B	Minasny et al. (2009);
		0.86	A	Viscarra Rossel et al. (2006)
	Vis–NIR	0.78	B	Sudduth et al. (2010);
		0.66–0.74	B	Mouazen et al. (2007);
		0.74	B	Metzger et al. (2024)
	NIR	0.03–0.52	C-E	Olfs et al. (2023);
		0.91	A	Chang et al. (2001);
		0.42	D	Metzger et al. (2024)
	Gamma-ray	0.20–0.76	B-E	Viscarra Rossel et al. (2007)
	XRF	0.27–0.38	D-E	Tavares et al. (2020)
Electrochemical	ISEs	0.93–0.96	A	Adamchuk et al., (2004, 2005, 2007);
		0.93	A	Sethuramasamyraja et al. (2008)
	ISFET	0.59	C	Artigas et al. (2001);
		0.62	B	Viscarra Rossel and Walter (2004);
		0.82	A	Viscarra Rossel et al. (2005)
Geoelectrical	EC	0.63–0.84	A-B	Schirrmann et al. (2011);
		0.58–0.98	A-C	Kweon et al. (2012)
Nitrate				
Spectral	UV-absorption	> 0.97	A	Shaw et al. (2013)
	MIR	> 0.98	A	Jahn et al. (2006);
		0,56	C	Cobo et al. (2010)
	Vis–NIR	0.19	E	Shibusawa et al. (2001)
	NIR	0.05	E	Olfs et al. (2023);
			C-D	Vikuk et al. (2024)
Electrochemical	ISEs	> 0.41	C	Adamchuk et al. (2005);
		> 0.98	A	Kim et al. (2007);
		0.94	A	Sibley et al. (2010);
		> 0.97	A	Shaw et al. (2013)
	ISFET	> 0.80	A	Artigas et al. (2001);
		> 0.90	A	Birrell and Hummel (2001)
Geoelectrical	EC	0.61	B	Sethuramasamyraja et al. (2008)

Table 2 (continued)

Sensor technology		R^2	Accuracy[(a)]	Reference(s)
plant-available Phosphate				
Spectral	MIR	0,49	C	Cobo et al. (2010);
		0.47	C	Du and Zhou (2009);
		0.84	A	Janik et al. (2009);
		0.20	D	Viscarra Rossel et al. (2006);
		0.38	D	Pätzold et al. (2020)
	Vis–NIR	0.69	B	Mouazen et al. (2007);
		0.78–0,93	A	Pätzold et al. (2020)
		0.92	A	Metzger et al. (2024)
	NIR	< 0.28	E	Olfs et al. (2023);
		0.47	C	Chang et al. (2001);
		0.90	A	Metzger et al. (2024)
	XRF	< 0.03	E	Tavares et al. (2020);
		0.46	C	Pelegrino et al. (2022);
		< 0.14	E	Andrade et al. (2020);
		0.29	D	Lima et al. (2019)
		0.03–0.90	A-E	Teixeira et al. (2018)
Electrochemical	ISEs	> 0.82	A	Kim et al., (2007, 2013)
	ISFET	0.99	A	Benslimane et al. (2023)
plant-available Potassium				
Spectral	MIR	0.82	A	Cobo et al. (2010);
		0.68	B	Du and Zhou (2009);
		0.53	C	Janik et al. (1998);
		0.18	E	Minasny et al. (2009)
	Vis–NIR	0.82	A	Mouazen et al. (2010);
		0.52	C	Metzger et al. (2024)
	NIR	< 0.38	C	Olfs et al. (2023);
		0.68	C	Chang et al. (2001);
		0.56	C	Metzger et al. (2024)
	Gamma-ray	0.50–0.90	A-C	Wong and Harper (1999)
	XRF	0.81–0.90	A	Tavares et al. (2020, 2023)
		0.04–0.12	E	Teixeira et al. (2018)
		0.81	A	Silva et al. (2021)
		0.58–0.67	B-C	Andrade et al. (2020)
Electrochemical	ISEs	> 0.61	B	Davenport & Jabro (2001);
		0.62	B	Adamchuk et al. (2005);
		0.50–0.54	C	Sethuramasamyraja et al. (2008);
		0.76	B	Kim et al. (2009)
	ISFET	0.81	A	Artigas et al. (2001)

The scale of accuracy (Chang et al., 2001; Kuang et al., 2012; Lorenz et al., 2023) is indicated by the letter A–E, as follows: A, excellent ($R^2 \geq 0.8$)—measurable with very high accuracy; B, good ($0.6 \geq R^2 < 0.8$)—measurable with high accuracy; C, approximate quantitative prediction ($0.4 \geq R^2 < 0.6$)—measurable with medium accuracy; D, distinguish between high and low ($0.2 \geq R^2 < 0.4$)—measurable with low accuracy; E, not usable ($R^2 < 0.02$)—measurable with very low accuracy.

Other abbreviations are as follows: R^2 coefficient of determination, *UV* spectral ultra-violet absorption, *EC* geoelectrical conductivity; *XRF* X-ray fluorescence, *ISEs* ion-selective electrodes, *ISFET* ion-selective field effect transistors

[a]Classification of accuracy into A, B, C, D, and E was based on the maximum number of publications confirming an accuracy category for a soil property

based, for example, on different soil texture or organic matter content. However, these parameters, such as texture and organic matter content, are very important variables for fertilizer recommendations because they are often used as co-factors for soil mapping to derive fertilizer management decisions for plant-available nutrients or liming.

Operation principles and sensor measurement techniques

Spectral sensors

Spectral sensors are a fast and cost-effective alternative to standard laboratory procedures for predicting soil properties (Ahmadi et al., 2021). They can be used in an invasive or non-invasive way. The principle is based on measurements of electromagnetic waves that are either emitted or reflected by the soil material (Meyer et al., 2019) depending on the specific soil characteristics (e.g., material composition, mineralogy). Statistical techniques are then employed to estimate the properties of the soil material based on the correlation between the reflected radiation and the soil properties of interest. Spectral measurements can be differentiated based on wavelength, channel bandwidth (ranging from narrow [10 nm] to broad [100 nm]), the number of channels (single-spectral, multi-spectral with fewer than 10 wave bands, and hyper-spectral with more than 10 wave bands), and presence of a light source (active or passive). The range of spectral application depends on the soil depth to be investigated, the soil surface conditions, and the soil properties to be determined (e.g. Nocita et al., 2015; Tilse et al., 2023).

The representativeness of measurements with respect to soil depth is an important aspect to consider. Electromagnetic waves penetrate the soil only to a few millimeters using active or reflection-based measurements in the gamma to infrared range (e.g. Huete & Escadafal, 1991; Mulla, 2013; Santos et al., 2023; Tilse et al., 2023). Increased wavelengths result in a higher penetration depth, but are still limited to a few centimeters. Visible range sensors can only capture the topmost layer of a few millimeters (Ahmadi et al., 2021), while radar-based microwaves can penetrate deeper into the soil up to several centimeters (Yuzugullu et al., 2020). On the other hand, gamma spectrometry, which employs geogenic background radiation, provides information on the soil at greater depths (Tavakoli et al., 2022) and interpretation in terms of soil or mineral properties (Egmond et al., 2010).

In order to obtain accurate results, it is essential to take into account the soil surface conditions at the point of measurement. Results may be less precise for rough or tilled soils, or soils with a high stone content, than for already leveled, recompacted soil surfaces (Nocita et al., 2015; Yuzugullu et al., 2020). To improve the interpretation of sensor results, other co-variables such as cropping system or tillage regime are important inputs. These aspects will be further discussed in Sect. 2.5. Spectroscopic analysis is a reliable method to determine the total nutrient content (Araújo et al., 2015), in addition to typical soil characteristics such as soil texture, moisture, and humus (Tabatabai et al., 2019; Tavares et al., 2020). Determining the total nutrient content is usually much simpler than determining the plant-available nutrient content because there are often clear features in the signal that make the predictions more accurate and robust for total element than for plant-available nutrients.Typically, a large share of nutrients is tightly bound to soil minerals (Ruehlmann et al., 2021) which makes total nutrient contents much larger than contents of plant-available nutrients.

According to Mulla (2013), shortwave (gamma, X-ray, and ultraviolet [UV]), visible (Vis-), NIR, MIR, micro, and radar wavelengths are utilized to determine soil properties. Using gamma-ray techniques, the potassium concentration (e.g., Wong & Harper, 1999) and pH in the soil can be predicted in the field (e.g., Viscarra Rossel et al., 2007). Moreover, gamma-ray techniques can provide information regarding the soil parent material and texture at a high spatial resolution. Radiometer counts are calibrated to ground truthing measurements and eventually to co-factors, but radiometer sensors are expensive and not widely used in precision agriculture.

Portable XRF sensors (Andrade et al., 2020; Janik et al., 1995; Silva et al., 2018, 2021; Stockmann et al., 2016; Weindorf & Chakraborty, 2016) and LIBS sensors (Erler et al., 2023; Tavakoli et al., 2022) are technologies used for nutrient measurements both in the laboratory and, recently, also in the field. The output data of XRF and LIBS sensors are related to the total content of elements present in the soil. However, recent studies (Pelegrino et al., 2022; Tavares, 2021; Tavares et al., 2023) have revealed that XRF data can likewise be used as a proxy for predicting available nutrients in tropical soils, which are highly weathered and possess oxides and kaolinite mineralogy, with very low amounts of macronutrients in their structure (Table 2). The application of XRF and LIBS is limited by spatial coverage and time needed for measurements. In order to obtain an accurate measurement, there is a need to prepare the soil sample in the field (e.g. by drying and/or sieving), which is hardly possible in the field. Other limitations for in-field nutrient measurements with LIBS, such es highly noisy data due to uncontrolled field conditions, are described by Hahn & Omenetto (2012). Although LIBS and XRF technologies are already commercially available, they are expensive, and their implementation may not have a direct agronomic benefit in precision agriculture (Viscarra Rossel & Lobsey, 2016).

UV absorption can identify nitrate concentrations (Table 2) via direct inference. However, nitrate absorption in the UV range is influenced by organic matter, which also absorbs at this wavelength. To reduce this influence, it is recommended that the soil extract is filtered or centrifuged (Shaw et al., 2013).

NIR spectroscopy (700—2500 nm) is a commonly used spectral technique in soil analysis, and is often combined with visible light (Vis–NIR; 400—700 nm) (e.g. Piccini et al., 2024). In the laboratory, typically, the soil samples are dried, sieved, and pressed into an analysis cup, which means that this approach requires a relatively simple sample preparation. In the field, Vis–NIR sensors are often used directly on the soil surface, or they are moved through the upper soil layer Vis–NIR systems used under field conditions are portable and do not require sample pretreatment, which makes them suitable for installation on soil probes. However, better results can be obtained under static in-situ field conditions (Kuang et al., 2012). Also, Vis–NIR technologies need site-specific calibration under controlled measurement conditions (Shibusawa, 2003, Sinfield et al., 2010).

Several studies have identified Vis–NIR spectroscopy as a powerful analytical tool for predicting soil chemical and physical characteristics, requiring only simple sample preparation and with no security issues. Vis–NIR spectroscopy seems to be suitable for a broad characterization of soil (Table 2) both in the laboratory and in the field (e.g. Pätzold et al., 2020; Piccini et al., 2024; Stenberg et al., 2010; Vibhute et al., 2020). For instance, in addition to the pH value, Vis–NIR can detect plant-available phosphorus reasonably well (coefficient of determination [R^2]: 0.09–0.80) (Mouazen et al., 2007), whereas for nitrate detection an R^2 value of only 0.19 was reported (Shibusawa et al., 2001). A review of the current status of mobile Vis–NIR measurement systems confirms the availability of several robust systems (e.g. Christy, 2008; Mouazen et al., 2005; Nawar & Mouazen, 2017; Shibusawa

et al., 2001; Stenberg et al., 2010). Mouazen et al. (2007) developed a prototype for a soil reflectance mapping unit equipped with a Vis–NIR spectrophotometer. This mobile system was calibrated for pH, plant-available phosphorus (paP), total nitrogen, and moisture in different soils in Belgium and northern France. The results of this study revealed that it was possibility to provide quantitative in-field data for pH and paP with mean error values of 27.5 and 5.1%, respectively. This system is technologically advanced, but it is rather expensive (Kuang et al., 2012).

On-the-go sensors produce often highly noisy data, due to uncontrolled field conditions such as varying soil moisture and ambient light conditions as well as vibrational disturbances of the sensor while moving (Hahn & Omenetto, 2012; Knadel et al., 2023; Tabatabai et al., 2019). According to Kuang et al. (2012), who compared spectral operation techniques for in-field use, static in-situ Vis–NIR methods provide greater accuracy than mobile Vis–NIR techniques. Changes in the distance between the sensor and the soil, the presence of stones and plant roots, noise from technical vibrations, and difficulties in aligning the geolocation of soil samples taken for validation with the corresponding spectra taken at the same position, are some of the contributing factors (Mouazen, 2009; Piccini et al., 2024).

MIR spectroscopy operates in the mid-infrared region of the electromagnetic spectrum (2 500–25 000 nm) and therefore it is sensitive to the fundamental wave modes of molecules that allow precise detection of chemical soil parameters (Ahmadi et al., 2021). Based on this ability, MIR can provide more accurate results than NIR spectroscopy under certain conditions (e.g., dry soil sample material [Kuang, et al., 2012]). MIR spectroscopy has been shown to be precisely ($R^2 > 0.98$) in measuring soil nitrate concentrations (Jahn et el., 2006), and it can measure soil pH with a good degree of accuracy (Minasny et al., 2009; Viscarra Rossel et al., 2006; Table 2). However, the accuracy of MIR for measuring paP tends to be lower than that for pH (Cobo et al., 2010), and Ahmadi et al. (2021) reported that MIR technology has a strong sensitivity to water absorption bands in the MIR spectrum, meaning that a varying soil moisture content will influence MIR measurements. The requirement for sample preparation (drying, crushing, sieving) is another limitation for field implementation of MIR technology in soil analysis (Kuang et al., 2012). However, since there are already examples of MIR technology being utilized in soil science, there is potential for its future application in agriculture (Leenen et al., 2019; Pätzold et al., 2020; Wehrle et al., 2022). In addition, there is already a database for MIR soil spectra available (Leenen et al., 2022).

Electrochemical sensors

Ion-selective electrodes (ISEs) are electrochemical sensors based on a potentiometric principle that directly measure the activity of selected ions in the soil solution. They are commonly utilized to measure specific ions in soils such as nitrate (NO_3^-), ammonium (NH_4^+), potassium (K^+), sodium (Na^+), and calcium (Ca^{2+}), as well as the pH value. ISEs are not yet widely used for the measurement of phosphate (Lobsey et al., 2010), but Kim et al. (2013) reported the possibility of detecting available phosphorus in less than 5 min (Table 2). Certain limitations to measurements with ISEs are imposed by the need for frequent calibration (Kim et al., 2006) and by their temperature sensitivity (Sethuramasamyraja et al., 2008). When using ISEs it is also a challenge to maintain good soil electrode / membrane contact in a field setting (Adamchuk et al., 2002). Moreover, ISEs

measurements require relatively high soil moisture (Adsett et al., 1999) and mechanical preparation (Adamchuk et al., 2005).

ISFET sensors (Bergveld, 1970) combine two well-established technologies: ISEs and field effect transistors (FET). These sensors operate on the principle of ion exchange, i.e., a selective membrane enables the passage of specific ions and generates an electrical signal proportional to the ion concentration. The measurement results are derived from obtained electrical voltage values.

Until now, the use of ISFET for soil measurement in precision agriculture has been limited. Recent studies, however, indicate that ISEs and ISFET have the potential for rapid in-field soil analysis (Adsett et al., 1999; Archbold et al., 2023; Artigas et al., 2001; Benslimane et al., 2023; Guerrero et al., 2021; Kashyap & Kumar, 2021; Kuang et al., 2012; Sethuramasamyraja et al., 2007; Tsukor et al., 2019). In the measurements, the ISFET offer several advantages over the traditional ISEs glass electrode, such as the ability of ISFET to be miniaturized on small chips and the resistance of ISFET to breakage (e.g., Beale et al., 2024). The durability, portability and fast response are further key advantages of ISFET (Beale et al., 2024; Guerrero et al., 2021; Viscarra Rossel et al., 2011), allowing direct measurements of soil chemical properties. The measurement with ISFET requires an ex-situ physical soil sample. There are different approaches of sample preparation, variations in extraction methods, variations in reaction time (Najdenko et al., 2023). Only a small amount of sample volume is needed to enable the detection of ion concentrations in soil extracts, making them well-suited for measuring plant-available soil nutrients. A challenge with ISFET, though, is that the extract must be filtered before measurement. Furthermore, other issues related to the application of electrochemical sensors are that detection may be disrupted by interference from other similar but undesired ions present in soil extracts (Archbold et al., 2023; Kim et al., 2007). ISFET sensors can also exhibit drift and a delayed response, leading to slower measurements of up to 5 min compared to spectral sensors (Sinfield et al., 2010). Measurements with ISFET may need a sensor rinsing between samples (Birrel and Hummel, 2001), à priori knowledge of soil type (Price et al., 2003), and, for rapid in-field use, improvement of the soil extraction system (Artigas et al., 2001). When using ISFET technology, the ISFET conditioning process is essential for first-time use or after prolonged dry storage. (Riedel et al., 2024).

Geoelectrical conductivity sensors

Soil electrical conductivity (EC) sensors are among the technologies that are moving precision agriculture from a concept to a potential tool for mapping soil properties directly or indirectly related to EC, such as water content, salinity, soil texture, and depth of clay layers (Sudduth et al., 2005). Currently, there are two types of machine-based geoelectric sensors available for EC measurements: invasive galvanically coupled electrode sensors (e.g. Veris Technologies; Geophilus) and non-invasive electrical impedance (EMI) sensors, such as Geonics' "EM 38", which acquires data at one or two soil depths, and Geoprospectors' "Topsoil Mapper", which acquires data at four different depths. Results from sensors based on EC are mainly influenced by the water and (in dry climates) salt content of the soil (Heege, 2013; Lorenz & Münchhoff, 2015). Furthermore, a reasonably strong correlation exists with the clay and organic matter contents in the soil (Meyer et al., 2019), which is needed for macronutrient fertilizer recommendation. This is a clear strength of EC measurements making them so valuable for precision agriculture. This kind of measurement is helpful for identifying site- or field-specific differences in the field, and is often used in

precision agriculture for field mapping as a tool for deciding where to take soil samples (Hinck et al., 2014; Lorenz et al., 2023). Because of the good correlation of the measured EC values with many soil properties, this method is frequently used as an auxiliary method to indirectly infer the soil pH and nutrient contents (Adamchuk et al., 2007; Kweon et al., 2012; Ruehlmann et al., 2021; Schirrmann et al., 2011; Table 2).

Operating methods and practical tools for in-field soil analysis

Soil analysis sensors can be operated while being mounted on a vehicle (machine-based), they can be carried by a person from one location to another (handheld) or, they can be kept in a static mode at a fixed site (Table 1). Static in-field approaches use sensors that record measurements under stationary conditions (i.e., field sensors localized to the same position of field, [Gerken, 2023; Kim et al., 2013). Handheld sensors are transportable devices or tools, such as probes or tubes that can be easily carried by hand to make a single observation at a specific point in time (in- or ex-situ) (e.g. Tavares et al., 2023; VIkuk et al., 2024). In contrast to static technologies, mobile systems determine soil properties while crossing the field either in a stop-and-go or an on-the-go mode, taking measurements directly (physical process) or indirectly (proxy process). Machine-based technologies can be attached to a vehicle or be self-propelled. The latest techniques for precision agriculture serve as independent self-contained field laboratories that allow real-time soil analysis directly on site. A common concept of so-called "mobile field laboratories" (i.e., Lab-on-chip, Field Labs) is to analyze soil samples instantly in the field and transfer the raw data to a server via internet for instant calculation of the results (e.g. Riedel et al., 2022; Tavakoli et al., 2022; Vikuk et al., 2024). This facilitates novel applications, such as comparing current analyses results with already existing ones or combining existing spectral libraries with in-situ measurements using algorithms (Knadel et al., 2023; Piccini et al., 2024; Viscarra Rossel & Behrens, 2010) and, if required, immediate resampling in the case of unclear results. These new applications enable users to improve their information on soil nutrients in a timely manner, often including cloud-based data analysis and transfer of recommendations to mobile devices.

Case studies for in-field plant-available nutrient analysis

At the case-study level, several technologies are already being used under practical farming conditions, although not always independently validated by research. Initial research on in-field nutrient sensing focused on soil pH (Viscarra Rossel & McBratney, 1997; Viscarra Rossel et al., 2005) and nitrate concentration (Birrell & Hummel, 1997), while more recent research on proximal soil sensing for soil pH has indicated significant success (e.g. Adamchuk et al., 2007; Borchert et al., 2011b; Olfs et al., 2010; Vogel et al., 2023). This is because the soil pH is relatively easy to measure and strongly correlates with soil properties that affect or buffer pH, such as soil minerals or organic matter. Considerable attention has also been paid to measuring the nitrate concentration, which emphasizes the need for site-specific management and variable-rate application of nitrogen in precision farming (Lobsey et al., 2010).

Prototypes have been validated based on laboratory and field experiments to evaluate the accuracy and precision of in-field measurements. These include studies by Ahmadi et al. (2021), Archbold et al. (2023), Kim et al. (2013), and Kuang et al. (2012). For

example, a static in-situ sensor system that can simultaneously analyze nitrate, paP, and plant-available potassium (paK) via ISEs in a soil–water extract was reported by Kim et al., (2007, 2009, 2013). The results obtained with this ISEs technique were consistent with the reference standard methods, with R^2 values of 0.92 (for nitrate), 0.82 (for phosphorus), and 0.97 (for potassium). The research project "Soilmonitor" aims to develop a static, battery-powered, invasive mini-laboratory for in-field measurement that will automatically analyze the soil solution (gained by ceramic suction filtration) for nitrate, phosphorus, and pH using an optoelectronic detection chip with organic light-emitting diodes (Gerken, 2023). The obtained analytical data will be transmitted in real-time through a wireless connection and combined with a soil moisture sensor.

Recently, a study by Tavares (2021) presented an alternative non-invasive in-situ device in the form of a spectral and easy-to-use portable XRF tube that enables the simultaneous detection of total nutrient contents and derivation of plant-available phosphorus and potassium (Tables 1 & 2).

Mobile platforms, mounted on vehicles or tractors, are used to collect data on-the-go (Archbold et al., 2023; Tsukor et al., 2019; Veris Technologies, 2003). These machine-based platforms may integrate one or more sensors to detect several parameters, the aim being to map the spatial variability of various soil characteristics (Table 1). Automated sensors enable high-density sampling (Adamchuk et al., 2004; Adsett et al., 1999; Viscarra Rossel & Walter, 2004). According to Archbold et al. (2023), these on-the-go soil nutrient measurements are the best approach for implementation in precision agriculture.

Several platforms based on the on-the-go principle have been introduced by Veris Technologies using automatic soil sampling systems for rapid in-field pH measurements. For example, Agricon GmbH used the existing Veris Technologies system with a pair of antimony pH electrodes, wherein the pH is measured by direct contact of the soil sample with the two electrodes (Decker et al., 2019). The pH is measured in untreated, naturally moist soil material, i.e., no solution is added to the soil before it comes into contact with the electrodes. The pH is then calculated from the average voltage outputs of the two electrodes. This system has shown good success on the basis of its agreement with reference data (Decker et al., 2019). In addition, Veris Technologies Soil pH Manager systems provide highly accurate in-field measurements of soil pH compared to measured references (Borchert et al., 2011a; Ruehlmann et al., 2023; Table 1).

A number of in-situ real-time soil analysis systems for the electrochemical detection of nitrate have been reported to show a very high level of accuracy. They are therefore very well-known and they can be based on the use of ISEs (e.g., Sibley et al., 2008, 2010) or, a combination of ISFET systems and flow injection analysis (e.g. Birrell & Hummel, 2001). A mobile automated system for rapid in-situ pH measurements using ISEs has recently been reported by Vogel et al. (2023).

Currently available static and mobile systems can determine many soil parameters relevant for fertilizer recommendations, such as pH, plant-available soil nutrients, soil texture, and organic matter, through contact with field-moist soil in the field. However, measuring different nutrients simultaneously is often complicated or requires additional sensor technologies for subsequent fusion of data (e.g. Javadi et al., 2021; see also Sect. 2.5). If plant-available nutrients need to be determined, the accuracy might be improved using a stepwise soil preparation procedure (Najdenko et al., 2023) so that extraction and filtration of soil samples can be integrated in a way similar to that used in the laboratory. The degree of accuracy might also be improved by combining different sensors into a multi-sensor platform (MSP) for data fusion.

Combinations of sensors

The combination of soil sensors leads to data fusion of essential soil properties such as soil pH, organic matter, texture, moisture and nutrients. Soil properties strongly associated with the soil composition are often simpler to measure than those of individual components. For instance, it is easier to determine the total nutrient content than the plant-available nutrient content (Araújo et al., 2015; Kodaira & Shibusawa, 2013; Viscarra Rossel et al., 2006).

Currently, gamma-ray and X-ray, as well as some spectral and electrochemical sensors, have the greatest potential for use in agriculture, as each of them can directly or indirectly measure or estimate multiple soil properties. Combining these sensors with existing mobile techniques might provide a soil analysis device for mapping plant-available nutrients in-field. By fusing data from multiple sensors, non-invasive real-time analysis of the soil nutrient status is possible (Javadi et al., 2021; Molin & Tavares, 2019; Tavares et al., 2020; Xu et al., 2019), and can be integrated into so-called "Smart Farming" systems, where they are connected to a central control unit or network to build a decision support system for the farmer. The sensor fusion approach combines the detection methods or resources to fuse, recalculate, or combine data. For spectral sensors, the use of more channels (e.g. multi or hyper-spectral) seems to be clearly better, although a smaller number of channels or a larger channel bandwidth can also lead to practically applicable prediction results (Metzger et al., 2024). For electrochemical sensors, it is often necessary to establish frameworks that allow their data to be fused with those of other sensors, enable telecommunication, and promote practical applications (Archbold et al., 2019). In the literature, there are a few reviews on the use of ISEs or ISFET, among other sensor technologies, for agricultural and environmental applications (e.g., Archbold et al., 2023; Guerrero et al., 2021; Kashyap & Kumar, 2021; Lehmann & Grisel, 2014; Nadporozhskaya et al., 2022) and in precision agriculture (Yin et al., 2021).

The Veris Technologies MSP (Veris Technologies, 2003) (Table 1) combines the technologies of Vis–NIR spectrometry, EC, and pH electrodes (Kweon et al., 2012). The Veris MSP has been used to simultaneously detect on-the-go nutrient levels of nitrate and potassium and the soil pH (Sethuramasamyraja et al., 2008), or pH only (Ruehlmann et al., 2021; Schirrmann et al., 2011), using ISEs. These pH electrodes make direct contact with the soil without chemical extraction and have yielded R^2 values of 0.61 (Sethuramasamyraja et al., 2008), 0.62 (Schirrmann et al., 2011), 0.36–0.79 (Olfs et al., 2010), and 0.93 (Ruehlmann et al., 2021) compared to reference measurements. Meanwhile, an automated on-the-go Veris Technologies system, which was developed and described by Adamchuk et al. (2005), can prepare fresh soil samples similar to a standard wet-chemical laboratory method, i.e., the soil is extracted with water and the levels of NO_3^- and K^+, as well as the soil pH, are immediately measured using ISEs (Table 1). After combining the ISEs and EC techniques, the results showed R^2 values of 0.93–0.96 for pH, 0.61–0.62 for potassium, and 0.41–0.51 for nitrate, compared to reference measurements.

The on-the-go multisensory platform called RapidMapper (Table 1) was developed as part of the "Intelligence for Soil I4S" project in Germany (Pätzold et al., 2020), which focuses on site-specific management of the main nutrients of nitrogen, phosphorus, and potassium, as well as on the soil pH, texture and organic matter. The platform utilizes established measurement techniques like EC, pH-potentiometry, and gamma-ray spectroscopy, alongside technologies such as XRF, NIR, and LIBS. By collecting sensor data, the platform provides information on soil properties such as nutrient and water availability via a decision support system (Tavakoli et al., 2022).

The research project "soil2data" (Germany) (Table 1) has recently introduced a new mobile field laboratory concept for in-field analysis (FieldLab). It utilizes a new express soil analysis method, based on a wet-chemical approach including the soil preparation procedure in two successive extraction steps, as developed by Najdenko et al. (2023) and Riedel et al. (2022). ISFET multisensor measuring technology is used, which simultaneously measures the pH and ionic concentrations (NO_3^-, $H_2PO_4^-$, K^+). The concept includes the in-field measured analysis results being sent to a cloud server and fused with other relevant soil properties to calculate individual fertilizer recommendations (Hinck et al., 2018). According to Najdenko et al. (2023), the results align with reference standard methods for NO_3^- ($R^2 = 0.95$), $H_2PO_4^-$ ($R^2 = 0.70$), K^+ ($R^2 = 0.81$), and pH ($R^2 = 0.91$).

As an alternative to conventional soil analysis methods, the start-up company Stenon (Table 1) has developed a fast-real-time direct soil measurement system called "FarmLab " with a combination of spectral UV/Vis-/NIR and electrical impedance sensor measurements that works with direct soil contact (Stenon, 2021; Vikuk et al., 2024). Sensor data on nutrient status are derived using an artificial intelligence (AI) machine learning system. For this soil analysis method, Olfs et al. (2023) compiled several independent test series with practical relevance, wherein the R^2 values ranged from 0.00 to 0.28 for phosphorus, 0.00 to 0.38 for potassium, and 0.03 to 0.77 for pH, compared to reference measurements.

ChrysaLab (Canada) (Table 1) is a handheld spectral soil analysis tool for determining soil nutrients (nitrate, potassium, phosphorus) and pH that combines three measurement technologies: Vis–NIR, spectral attenuated total reflection (known as ATR infrared spectroscopy which enables samples to be examined directly in the solid or liquid state without further preparation), and an AI machine learning system (Chrysalabs, 2023). However, there are still no independent studies on the reliability of this measuring system available.

The combination of soil sensors provides options for data fusion of essential soil properties such as soil pH, soil organic matter, texture, moisture and nutrients, which are key prerequisites for the calculation of fertilizer recommendations in many countries. Moreover, knowledge of these soil properties is highly relevant for site-specific cultivation measures such as seeding / seeding density or ploughing depth, as well as for irrigation and site-specific liming.

An exciting aspect of the sensor combination is that it provides continuous, real-time monitoring of the soil, allowing changes and potential problems to be identified more rapidly. It also allows multiple fields to be monitored simultaneously, improving labor and resource efficiency. Using a combination of sensors can lead to better and knowledge-based decisions, resulting in improved soil management, higher crop yields, improved soil fertility and reduced environmental impact. The use of soil sensors and data fusion can foster collaboration between researchers, farmers and other stakeholders by encouraging the sharing of data and expertise. This can contribute to advancing soil management and agricultural research. Finally, the combination of soil sensors and data fusion enables the development of autonomous and semi-autonomous systems that can manage various agricultural tasks such as soil sampling, fertilization, and irrigation, saving time, effort, and resources for agricultural professionals.

Important terms of data assessment and management for in-field measurements

Data acquisition systems are used to collect data for research, analysis, validation and verification, and quality testing in-field. One of the most important specifications of a data management system is the data acquisition sampling frequency, i.e. the data sampling rate per unit of time. This term is important for determining the request-response signals of sensors, and represents the pre-established soil sample treatment and analysis reaction time of the soil sample in-field. The accuracy of the sensor measurements and the reliability of the results is represented by the interaction of the trueness (mean) and precision (standard deviation) of the measurements (Menditto et al., 2007; Thode, 2023). Reliability is the quality of being dependable, trustworthy, or performing consistently well and requires that things work as expected under normal, known circumstances (Piccini et al., 2024). Robustness is the ability to perform without failure under a wide range of possible conditions (Jones, 2021).

New in-field methods provide measurement results that are calibrated against a corresponding reference method (Lorenz et al., 2023; Steinfurth et al., 2021), and the goal of the in-field method is to match the laboratory reference, which is usually a wet-chemical test method measured with the appropriate laboratory equipment, as closely as possible. Many conventional test methods for plant available nutrient are country-specific and therefore differ from each other (e.g. Higgins et al., 2023), which leads to a large range of different preparation and extraction procedures. Jordan-Meille et al. (2012) compared several methods for phosphorus analysis and the resulting phosphorus fertilizer recommendations across Europe and found great differences in the actual results of the laboratory soil tests, as well as the interpretation of the laboratory data for providing the final phosphorus fertilizer recommendation to the farmer (Higgins et al., 2023).

It has to be kept in mind that the results from rapid in-field methods cannot be transferred "one-to-one" to another analysis method or another advisory procedure. In any case, it is necessary to check whether there is a good correlation between the analysis results of two different test methods and whether there are any deviations between the absolute values (Najdenko et al., 2023; Steinfurth et al., 2021). If the correlation is high, a factor might be used to convert the analysis result so that a well-established advisory procedure can be applied.

In this context, calibration can be defined as the development of a data model to determine the best possible agreement between the measurement results of an in-field method and the test results of a reference laboratory. The calibration dataset should cover the full range of possible analytical results of the standard method, and the accuracy of the in-field measurements depends on the strength of the correlation and the material measured during calibration. Such a model is usually developed by, or on behalf of, the sensor manufacturer. Calibration is also used when a measurement method is adapted to specific site conditions, under which circumstances it does not necessarily involve the sensor manufacturer. Many approaches to rapid in-field soil analysis are successful in calibration, which is critical to the adaptation of a new method, but then fail in the next essential step, which is validation. Validation verifies that the calibration matches the promised quality of the measurement results. This is performed on samples that have not been part of the calibration, meaning that an independent set of samples is needed that covers the measurement range of the calibration and is representative of the application range of the method. Such a validation is usually performed by an independent organization. The validation proves that

the results are comparable with the reference values, which is very important for fertilizer recommendations.

Quality management is used to ensure the reliability of the measurement results even during operation of the sensor system, as the accuracy and reproducibility of the sensor can be affected over time, e.g. by possible sensor drift, sensor aging, variation in day-time temperature, as well as soil conditions like soil moisture, soil roughness, aggregation, stone content, and coarse organic residues (e.g. Piccini et al., 2024). Among other things, this requires regular functional testing. Verification is a critical component of quality management, helping to ensure that measurement procedures fulfil the desired quality levels, and focusing specifically on assessing and confirming compliance with predetermined requirements or standards.

Concluding remarks

Precision agriculture has developed rapidly in recent years, introducing new opportunities for in-field nutrient analysis. Soil sensing requires massive data collection for real-time soil analysis. In-field methods rely on the rapid availability of soil data to assess the distribution of soil properties and associated soil nutrients in the field. Preparation of soil samples for analysis is often required. Many field measurements are performed without soil preparation, but with subsequent direct data processing. In addition to individual measurements, almost all in-field methods require subsequent external data processing, taking into account laboratory references and field-specific on-site calibrations, typically done during initiation of each in-field measurement series. Using an approach of complementary soil sensing techniques with multi-sensor data fusion provides an opportunity to obtain more information about soil properties and soil nutrient status. Data fusion of multiple soil sensors can improve the quality of the obtained soil information data and provide a greater range of soil information that cannot be obtained with single sensors. Furthermore, machine learning technologies and other AI methods are highly promising approaches to achieve greater accuracy in the relationships between sensor signals, soil properties and nutrient status. Future sensors may be based on satellites, unmanned aerial vehicles, tractors, or attached to mobile robots to record soil properties and the soil nutrient status, which are important for fertilizer management decisions. Soil sensors need to be robust, they should be powered by renewable energy sources, and able to transmit information through wireless networks for data processing and generation of automated fertilizer management recommendations at high spatial resolution. A future challenge will be the development of precision farming approaches that provide customized management of inputs for individual crops. To accelerate broader adoption of rapid in-field soil analyses in farming practice, farmer education, and certification of in-field methods as official soil analysis methods are essential. Also, there is a need to assess technical limitations of these rapid in-field systems across a broader range of soils in order to avoid improper usage and declining confidence by farmers or contractors.

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Declarations

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